

Lab3-2

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4.5

n	$f(n)$
1	6
2	7
3	6
4	7
5	6

a. f is not one-to-one

b. f is not onto

c. f is not correspondence

n	$g(n)$
1	10
2	9
3	8
4	7
5	6

d. g is one-to-one

e. g is onto

f. g is correspondence

1/4

4.6

Assuming B is countable. So we can list all sequences:

$$S_1 = s_{11} s_{12} s_{13} \dots \quad s_{ij} \in \{0, 1\}$$

$$S_2 = s_{21} s_{22} s_{23} \dots$$

$$S_3 = s_{31} s_{32} s_{33} \dots$$

$\vdots$

we set $t = t_1 t_2 t_3 \dots$, and $t_n = 1 - s_{nn}$.

For each element in S set, $t_n \neq s_{nn}$, so $t \neq S_n$.

t is not in S .

But t is in B and B is countable.

So get contradiction.

B is uncountable.

$$4.7 \quad T = \{(i, j, k) \mid i, j, k \in \mathbb{N}\}.$$

For example,

$$(1, 1) \Rightarrow 1+1=2$$

$$(1, 2), (2, 1) \Rightarrow 1+2=3$$

$$(2, 2), (1, 3), (3, 1) \Rightarrow 1+3=2+2=4$$

$$m+n \rightarrow \text{pairs} \Rightarrow m+n-1$$

$$1+2+3+\dots+(m+n-2) = \frac{(m+n-2)(m+n-1)}{2}$$

f is function for order number $\in \mathbb{N}$ mapping to each pair.

$$f(m, n) = \frac{(m+n-2)(m+n-1)}{2} + m$$

So $\mathbb{N} \times \mathbb{N}$ is countable.

So $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N}$ is also countable.